

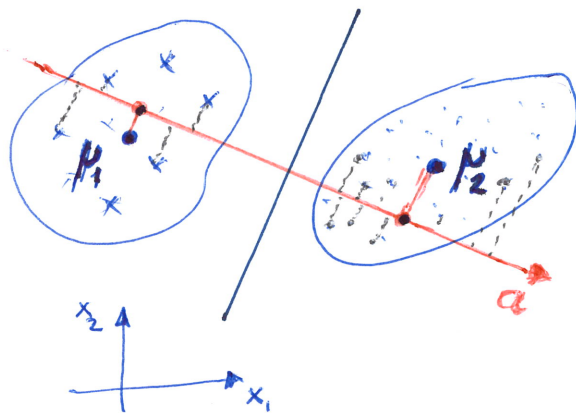
Chap 12 & 13: ML and NN

Exercise 1

Let x be an observation;

$a^T x$: projection of x onto a

This leads to two distributions



If x is a RV with mean μ , the

variance of the projection onto a is $E(a^T(x-\mu))^2 = E[a^T(x-\mu)a^T(x-\mu)]$
 $= E[a^T(x-\mu)(x-\mu)^T a] = a^T E[(x-\mu)(x-\mu)^T] a$
 $= a^T \Sigma a$

where Σ is the covariance matrix of the RV x .

For two classes: $x_1, \dots, x_{n_1}$, and $y_1, \dots, y_{n_2}$, the projection onto a are

$x_1^T a, \dots, x_{n_1}^T a$, and $y_1^T a, \dots, y_{n_2}^T a$, their respective means are

$\mu_1^T a$ and $\mu_2^T a$, and their variances are $a^T S_1 a$ and $a^T S_2 a$,
 (S_1 : covariance matrix of group 1, and similarly for group 2).

The within-group variances;

are: $a^T S_1 a$ and $a^T S_2 a$, and the total within-group variance is $a^T (S_1 + S_2) a$.

The between-group variance is the distance between the obtained averages: $[a^T (\mu_1 - \mu_2)]^2$.

Fisher LDF maximizes: $L(a) = \frac{[a^T (\mu_1 - \mu_2)]^2}{a^T (S_1 + S_2) a}$; the optimum a is obtained by solving $\nabla L(a) = 0$; that is:

$$\left[a^T (\mu_1 - \mu_2) (\mu_1 - \mu_2)^T \right] a^T (S_1 + S_2) a - \left(a^T (\mu_1 - \mu_2) \right)^2 (S_1 + S_2) a = 0$$

$$(a^T (S_1 + S_2) a) (\mu_1 - \mu_2) = (a^T (\mu_1 - \mu_2)) (S_1 + S_2) a \quad \text{or}$$

$$(S_1 + S_2) a = \alpha (\mu_1 - \mu_2) \Rightarrow \boxed{a = \alpha (S_1 + S_2)^{-1} (\mu_1 - \mu_2)}$$

The hyperplane is orthogonal to a .

Exercise 2

k-means is a unsupervised learning algorithm requires the parameter k ; the number of clusters. It clusters the data into k clusters/groups. It is efficient, and is easy to understand and implement.

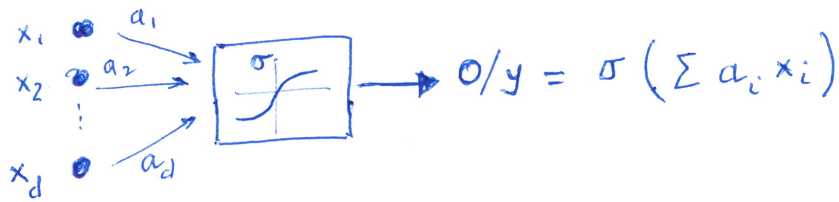
Hierarchical clustering produces a tree-like structure of the data (dendrogram) that we can visualize. And we can cut the dendrogram at any chosen level.

k-means depends on k ; a difficult parameter. It can also be sensitive to outliers.

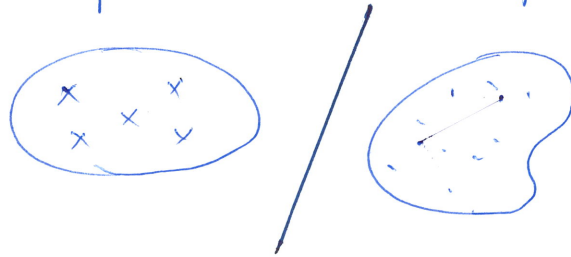
The structure of the dendrogram in hierarchical clustering depends on the dissimilarity used:

- complete
- single linkage
- average linkage
- centroid, etc...

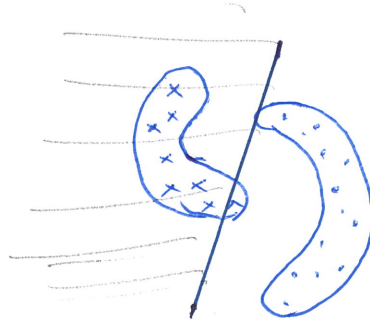
Exercise 3



This perceptron can separate linearly separable classes.

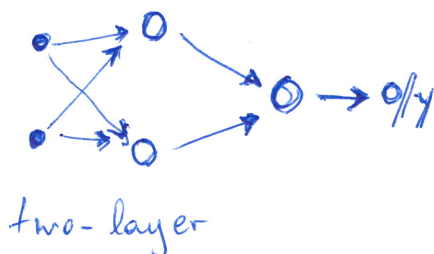


Decision
region =
half plane

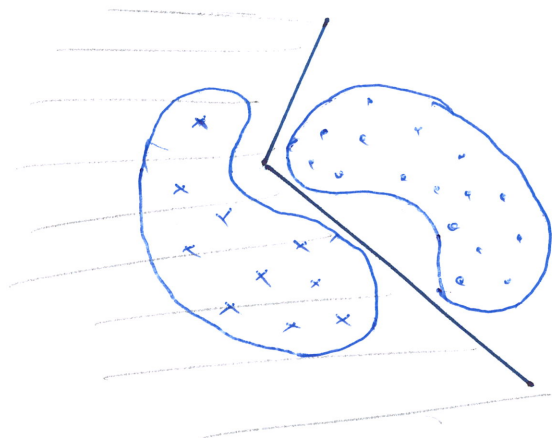


So this perceptron cannot do binary classification with nonlinear separation surface.

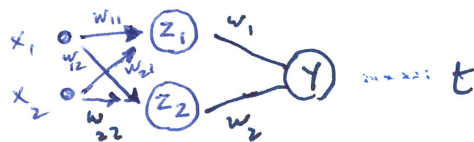
The answer lies in using two layers (at least) for nonlinear separation



two-layer



Exercise 4



$$Y = w_1 z_1 + w_2 z_2$$

$$\text{sigmoid: } \sigma(x) = (1 + e^{-x})^{-1}$$

We let $x^i = (x_1^i, x_2^i)$, $i = 1, \dots, n$, and

$$y_i = w_1 z_1^i + w_2 z_2^i = w_1 \sigma(w_{11} x_1^i + w_{21} x_2^i) + w_2 \sigma(w_{12} x_1^i + w_{22} x_2^i)$$

$$E = \frac{1}{2} \sum_{i=1}^n (y_i - t_i)^2, \quad \frac{\partial E}{\partial w} = \sum_i (y_i - t_i) \frac{\partial y_i}{\partial w}$$

where w can be any weight ($w_1, w_2, w_{11}, \dots$ etc).

$$\text{For any } i = 1, \dots, n; \quad \frac{\partial y_i}{\partial w_1} = \sigma(w_{11} x_1^i + w_{21} x_2^i) \text{ and } \frac{\partial y_i}{\partial w_2} = \sigma(w_{12} x_1^i + w_{22} x_2^i)$$

$$= \sigma(z_1^i) \quad = \sigma(z_2^i)$$

$$\frac{\partial y_i}{\partial w_{11}} = w_1 x_1^i \sigma(z_1^i) (1 - \sigma(z_1^i)); \quad \frac{\partial y_i}{\partial w_{21}} = w_1 x_2^i \sigma(z_1^i) (1 - \sigma(z_1^i))$$

$$\frac{\partial y_i}{\partial w_{12}} = w_2 x_1^i \sigma(z_2^i) (1 - \sigma(z_2^i)); \quad \frac{\partial y_i}{\partial w_{22}} = w_2 x_2^i \sigma(z_2^i) (1 - \sigma(z_2^i))$$

$$\text{Let } w = (w_1 \quad w_2 \quad w_{11} \quad w_{21} \quad w_{12} \quad w_{22})^T, \text{ and}$$

$$\nabla y_i = \left(\sigma(z_1^i) \quad \sigma(z_2^i) \quad w_1 x_1^i \sigma(z_1^i) (1 - \sigma(z_1^i)) \quad w_1 x_2^i \sigma(z_1^i) (1 - \sigma(z_1^i)) \right. \\ \left. w_2 x_1^i \sigma(z_2^i) (1 - \sigma(z_2^i)) \quad w_2 x_2^i \sigma(z_2^i) (1 - \sigma(z_2^i)) \right)^T$$

Then

$$\nabla E = \sum_{i=1}^n (y_i - t_i) \nabla y_i$$

And the updating rule is:

$$w_i \leftarrow w_i + \Delta w_i = w_i - \eta \frac{\partial E}{\partial w_i}, \quad 0 < \eta \leq 1: \text{learning rate}$$

that is

$$w \leftarrow w - \eta \nabla E$$